Water Quality C - Golden Gate SO 2021 - Water Quality Test - Golden Gate 2021 - 02-13-2021

Answer all questions to the best of your ability. Point values and tie-breakers are indicated by each question. There is no penalty for guessing!

Good luck! We believe in you :))

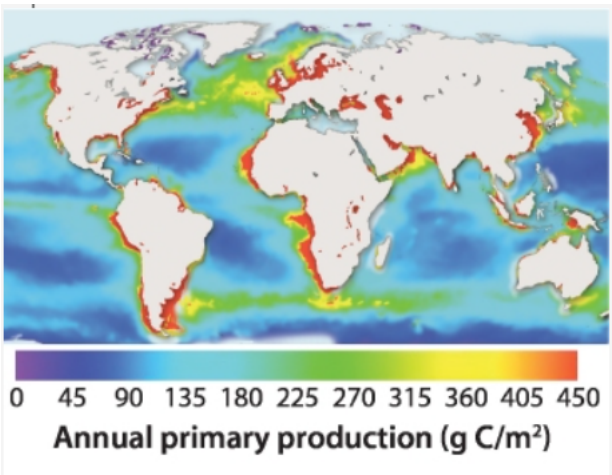
1. (1.00 pts) On my honor, I have not given, received, nor witnessed any unauthorized aid on this test.

- (●) A) Yes
- ( ) B) No

Part I: Marine & Estuary Ecology

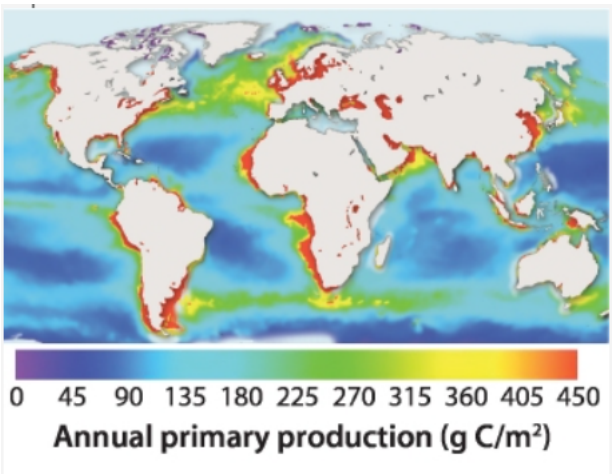
Questions 2-61

2. (1.00 pts)
This map illustrates annual carbon uptake due to primary production by primary producers in the ocean. Primary production is dependent on nitrogen and phosphorus. According to the map, where do higher levels of primary production occur?



- (●) A) Coastal Waters
- ( ) B) Open ocean
- ( ) C) Enclosed seas
- ( ) D) Between continents

3. (1.00 pts) Based on the previous question (and figure), what is the best explanation for the higher levels of primary production?



- A) Coastal waters receive more sunlight than open ocean, which is necessary for high levels of primary production
- B) Coastal waters have higher levels of nutrient-rich runoff due to human agricultural activities than the open ocean
- C) The open ocean has fewer primary consumers than the enclosed seas, so less primary production occurs in the open ocean.
- D) The enclosed seas are warmer than the open ocean, so primary production is more likely to occur there.

4. (1.00 pts) Which of the following statements regarding niches is true?

- A) Niches are dynamic and are affected by interactions between species
- B) Niches are dynamic and are not affected by interactions between species
- C) Niches are not dynamic and are affected by interactions between species
- D) Niches are not dynamic and are not affected by interactions between species

5. (1.00 pts) Which of the factors contribute most to intraspecific competition?

- A) Mutualistic relationship
- B) A species' fundamental niche and abiotic surroundings
- C) A species' rate of reproduction and the carrying capacity of the environment
- D) Parasitic relationship

6. (1.00 pts)
In the west coast of United States and Canada, kelp can grow more than 50 m tall. Sea otters prey on sea urchins, which anchor kelp to the seafloor. When sea otters were lost as a result of overhunting, the kelp nearly disappeared. Which of the three organisms (sea otters, sea urchins, or kelp) are the keystone species in this community?

- A) Sea otters
- B) Sea urchins
- C) Kelp
- D) Sea otters and sea urchins

7. (1.00 pts) A 1998 study found Alaskan kelp forests were in decline due to change in orca behavior. What do you think most likely drove orcas to prey upon sea otters?

- A) Orcas have always eaten sea otters, their populations were just growing

B) Orcas evolved to hunt sea otters because they are more nutritious than fish

C) Declining whale and seal populations caused a decline in preferred prey of Alaskan orcas

D) Climate change has caused orcas to move south from the Arctic into Alaskan waters where they began hunting sea otters

8. (1.00 pts) The kelp forest, sea otter, and sea urchin example shows what kind of trophic cascade?

A) Top-down cascade

B) Bottom-up cascade

C) Subsidy cascade

D) Equilibrium cascade

9. (1.00 pts) What kind of estuary occurs when the mouth of a river flows directly into salt water?

A) Fjord

B) Vertically stratified estuary

C) Freshwater estuary

D) Salt-wedge estuary

10. (1.00 pts) Diatoms and dinoflagellates are... (select all that apply)

(Mark **ALL** correct answers)

A) Phytoplankton

B) Zooplankton

C) Cyanobacteria

D) Protosomes

E) Protists

11. (1.00 pts) What is the uppermost zone of the ocean?

A) Mesopelagic

B) Abyssopelagic

C) Epipelagic

D) Hadopelagic

12. (1.00 pts) Which of the following is NOT a measure of turbidity in fluids? (Select all that apply)

(Mark **ALL** correct answers)

A) Jackson Turbidity Units

B) Nephelometric Turbidity Units

C) Formazin Nephelometric Units

D) Total Dissolved Solids

13. (1.00 pts) The dissolved oxygen in estuarine waters are generally as its lowest levels during the late summer months. Which of the following is the most likely reason for this?

○ A)
Oxygen is supplied primarily to estuarine waters via diffusion of atmospheric oxygen into the water, and the rate of mixing of surface waters and wind is lowest during the summer.

○ B) The photosynthesis by phytoplankton is lowest in the summer because they are in lowest abundance during the summer.

◉ C) The solubility of oxygen decreases with increasing temperature, which is why the levels are the lowest during the summer.

○ D) Oxygen is removed from the water by aerobic respiration, which is at its highest during the summer.

14. (1.00 pts) Which of the following is an example of nekton?

○ A) Phytoplankton

○ B) Copepod

○ C) Holoplankton

◉ D) Decapod

15. (1.00 pts) What is the carbon source for marine autotrophic organisms?

◉ A) Inorganic carbon

○ B) Phytoplankton

○ C) Dissolved organic carbon

○ D) Particulate organic carbon

16. (1.00 pts) What are the methods of electrodialysis, reverse osmosis, and ion exchanged used for?

○ A) Determining water salinity

○ B) Determining residence time

○ C) Biological decontamination

◉ D) Desalination

17. (1.00 pts)
If female lionfish don't reproduce until they are about ~12-14cm (many of the individuals remain juveniles in the population), how much slower body growth rates affect population growth?

○ A) Population growth rates will remain low near carrying capacity

○ B) Population growth rates will be high because when they reproduce the population will explode in size showing exponential growth

◉ C) Cannot determine the population growth based on the information given

18. (1.00 pts) How is organic matter produced in the ocean classified?

○ A) Allochthonous

◉ B) Autochthonous

○ C) Terrigenous

○ D) Hydrogenous

19. (1.00 pts) Where does primary production by marine plants occur?

A) Occurs at all depths
B) Is limited to the upper 200 meters of the sea
C) Is limited to the aphotic zone
D) Occurs below the compensation depth only

20. (1.00 pts) Which of the following is the cause of red tides?

A) Planktonic bloom of red-colored microorganisms
B) Excess of iron compounds in the water
C) Photosynthesis, combination of carbon and water
D) Organic dyes emitted by nudibranches and neustonic organisms

21. (1.00 pts) Which of the following is an example of a neritic environment?

A) An estuary
B) The sedimentary surface of the seafloor
C) Coastal ocean waters
D) The surface waters of the Great Lakes

22. (1.00 pts) Which of the following is NOT a source of particulate organic matter in the deep sea? (Select all that apply)

(Mark **ALL** correct answers)
A) Planktonic material
B) Marine macrophyte detritus
C) Terrigenous matter
D) Agricultural runoff

23. (1.00 pts) What does TOTO stand for (hint: in the Bahamas)?

A) Total oxygen to ocean
B) Tongue of the ocean
C) Thermal origin terminal order
D) Typhonian order templi orientis

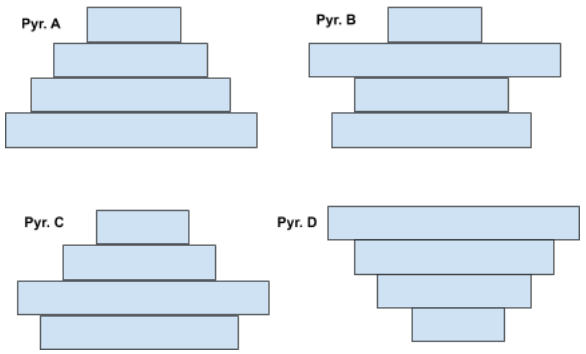
24. (1.00 pts) What is the name of the benthic organisms that live in the sediment?

A) Epibenthos
B) Mobile epibenthos
C) Infauna
D) Cladocera

25. (1.00 pts) What zone of the ocean are coral reefs classified in?

A) Oceanic zone
B) Neritic zone
C) Photic zone
D) Littoral zone

26. (1.00 pts)
(Tie-breaker) Which of the following trophic pyramids of biomass best describes aquatic ecosystems? - Note the bottommost layer represents the biomass of primary producers while the topmost represents the highest level of consumers.

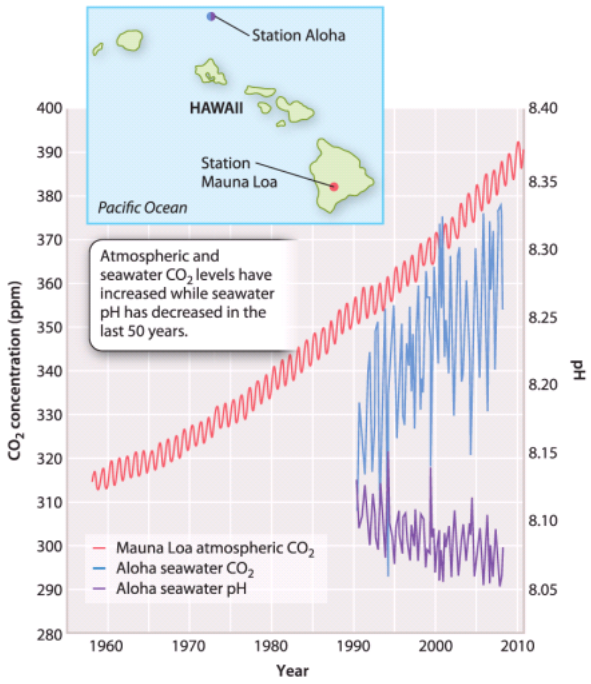


A) Pyramid A
B) Pyramid B
C) Pyramid C
D) Pyramid D

27. (1.00 pts) Hydrothermal vents generally expel fluids containing...

A) Hydrogen, hydrogen sulfide, and methane
B) Carbon dioxide, hydrogen sulfide, and oxygen
C) Carbon dioxide, methane, and ammonia
D) Hydrogen, oxygen, and nitrogen oxide

28. (1.00 pts) What phenomenon does the following graph illustrate?



- A) Bioaccumulation
- B) Coral bleaching
- C) Eutrophication
- D) Ocean acidification

29. (1.00 pts)
(Tie-breaker) You are trying to determine the NTU of an original sample, but the sample's NTU is too high. In order to determine the NTU, you decide to dilute the sample. Yo add 10mL of turbidity-free water to 1mL of sample. The diluted sample has a turbidity of 21 units. Calculate the turbidity of the original sample.

- A) 21 NTU
- B) 210 NTU
- C) 231 NTU
- D) 207 NTU

30. (1.00 pts) Which of the following is the difference in height between high and low tides?

- A) Tidal height
- B) Tidal range
- C) Tidal incidence
- D) Tidal current

31. (1.00 pts)
An industrial plant discharges 100 cubic ft/sec (cfs) of fluent with 10 ppt of toxic material into an estuary. The minimum tributary inflow upstream of the discharge is 1000cfs. The salinity at the discharge point is 21 ppt and the ocean salinity is 31 ppt. Estimate the average concentration of the toxic material in the estuary at the discharge point.

- A) 0.39 ppt
- B) 0.31 ppt

C) 0.19 ppt

D) 0.29 ppt

32. (1.00 pts)
(Tie-breaker) The water volume of an estuary is 1 x 10^6 m^3. The amount of water added from river flow is 200 m^3/s and from groundwater is 1 m^3/s. The amount of water that evaporates is 0.5 m^3/s. Calculate the turnover time for this estuary.

A) 1382 seconds

B) 3923 seconds

C) 4987 seconds

D) 6209 seconds

33. (1.00 pts) Which of the following does not affect significantly water circulation within a estuary? (Select all that apply)

(Mark **ALL** correct answers)

A) Tidal currents

B) Wind

C) Local topography

D) Drainage

34. (1.00 pts) Estuaries have also been called...

A) Bays

B) Lagoons

C) Sloughs

D) All of the above

35. (1.00 pts) Which of the following is the biggest estuary in the United States?

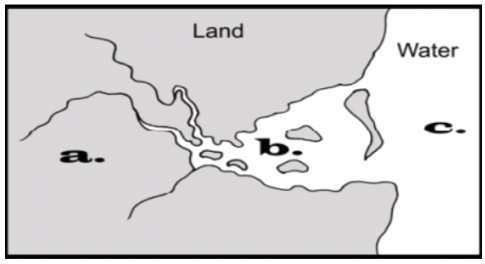
A) Chesapeake Bay

B) Chattahoochee River

C) Albemarle-Pamlico sound system

D) Hudson Bay

36. (1.00 pts) Which of the letters represents an estuary in the following diagram?



A) A

- [x] B) B
- [ ] C) C
- [ ] D) Combination of A and B

37. (1.00 pts) Which of the following is/are contaminant(s) of the San Francisco Bay?

- [ ] A) Mercury
- [ ] B) Polychlorinated biphenyls
- [ ] C) Organochlorine pesticides
- [x] D) All of the above

38. (1.00 pts) Which of the following is included in OSTDS?

- [ ] A) Sewers
- [ ] B) Pipes
- [x] C) Effluent pump
- [ ] D) Percolation pond

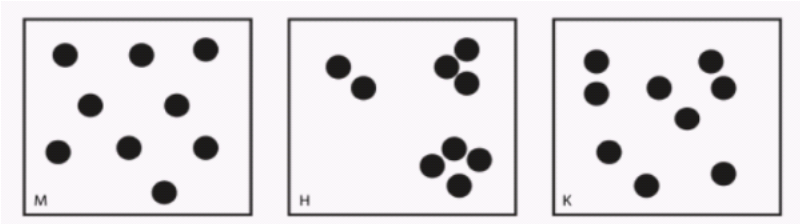
39. (1.00 pts)
A population of 10 pairs of butterfly fish colonize a new area with no predators and a number of resources which are enough to support a butterfly fish population of 2000 individuals. Each pair can have 30 offspring per year. For the first 2 years in a new area, the growth curve for this population will show:

- [ ] A) Exponential growth because only a few individuals will have offspring
- [x] B) Logistic growth because the colonizing population will reach carrying capacity within the first two years
- [ ] C) Exponential growth because the colonizing population size will remain below the carrying capacity for the first two years
- [ ] D) Logistic growth because they will reach maximum population size immediately

40. (1.00 pts)
There are two species of fish that can mate with each other, even though they look slightly different. According to what concept are these two species the same species?

- [ ] A) Ecological species concept
- [ ] B) Phylogenetic species concept
- [ ] C) Morphogenesis species concept
- [x] D) Biological species concept

41. (1.00 pts)
The figures below display 3 hypothetical populations. Each circle is an individual. Which figure depicts a situation in which individuals establish exclusive territories to secure access to resources?



- (●) A) Diagram M
- ( ) B) Diagram H
- ( ) C) Diagram K
- ( ) D) None of the above

42. (1.00 pts) Which of the following treatment options CANNOT be used to treat bacterial contamination?

- ( ) A) Chlorination
- ( ) B) Ultraviolet Light
- ( ) C) Distillation
- (●) D) Oxidizing filters

43. (1.00 pts) Minimum waste treatment requirements established by the Department are called…

- (●) A) Technology-Based Effluent Limitation
- ( ) B) Water Quality-Based Effluent Limitation
- ( ) C) Substantial modification
- ( ) D) Design capacity

44. (1.00 pts) In deeper waters, nutrients primarily come from:

- (●) A) Detritus sinking from more productive shallow waters
- ( ) B) Surface winds creating deep ocean currents
- ( ) C) Thermal vents
- ( ) D) Glacial ice melt

45. (1.00 pts) Which of the answer choices is directly responsible for depleting the supply of oxygen in bottom waters?

- ( ) A) Expanding fish and invertebrate populations
- ( ) B) Algal growth
- (●) C) Bacterial growth
- ( ) D) Nitrification

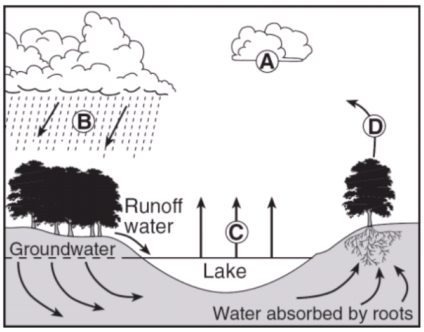
46. (1.00 pts) Which of the following explanations for the global increase in harmful algal blooms is the most plausible?

- ( ) A) Increased scientific awareness of toxic species
- ( ) B) Increased utilization of coastal waters for aquaculture
- ( ) C) Stimulation of plankton blooms by cultural eutrophication
- (●) D) All of the above

47. (1.00 pts) Which of the following water cycle processes return water vapor directly to the atmosphere:

A) Water retention and runoff
B) Freezing and precipitation
C) Infiltration and capillarity
D) Evaporation and transpiration

48. (1.00 pts) The letters A through D in the following image represents four water cycle processes:



Which of the following processes correctly corresponds with its letter:

A) A: Condensation | B: Precipitation | C: Evaporation | D: Transpiration
B) A: Evaporation | B: Condensation | C: Precipitation | D: Transpiration
C) A: Condensation | B: Precipitation | C: Transpiration | D: Evaporation
D) A: Transpiration | B: Precipitation | C: Evaporation | D: Condensation

49. (1.00 pts) Why is the volatile acid/alkalinity ratio important at a wastewater treatment plant:

A) It is commonly used to adjust the acidity in the chlorination facility.
B) It is an important indicator of a digester problem.
C) Used to control the pH of the influent in pretreatment facilities.
D) Helps maintain proper operation of the primary sedimentation process.

50. (1.00 pts) (Tie-breaker) When would you expect the lowest concentration of DO in a facultative pond:

A) 3 pm
B) Sunrise
C) Sunset
D) Noon

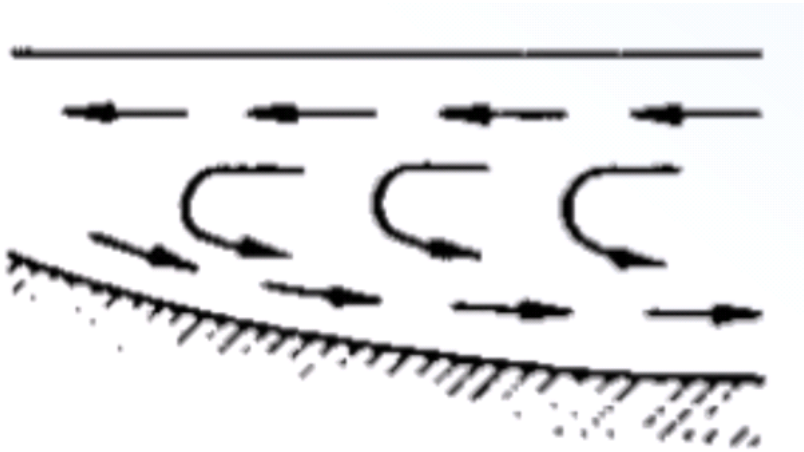
51. (1.00 pts) The deep ocean accounts for more than what percentage of Earth's carbon:

A) 5%
B) 20%
C) 50%
D) 65%

52. (1.00 pts) What estuary contains sill that results in stagnant bottom waters and is usually highly-stratified:

- A) Salt wedge estuary
- B) Partially mixed estuary
- C) Well-mixed estuary
- D) Fjord estuary

53. (1.00 pts) Which of the following types of circulation modes is depicted in the following image:



- A) Classical
- B) Reverse
- C) Three-layered
- D) Discharge

54. (1.00 pts) Which is an example of a semelparous species:

- A) Many pacific salmon
- B) Perennial plants
- C) Elephants
- D) None of the above

55. (1.00 pts) (Tie-breaker) Bat hedging has been proposed as an advantage to:

- A) Semelparity
- B) Iteroparity
- C) A and B
- D) None of the above

56. (1.00 pts) Which is the correct total hardness as $CaCO_3$ of a water sample in which Ca^{2+}= 143 mg/L and Mg^{2+} = 13 mg/L:

- A) 380 mg/L

B) 437 mg/L

C) 156 mg/L

D) 394 mg/L

57. (1.00 pts) The California Current is caused by:

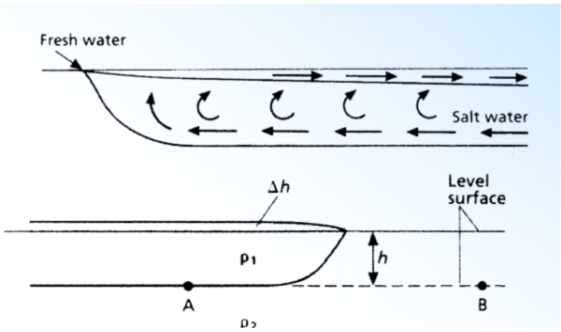
A) Thermohaline circulation

B) Gyre circulation

C) Coastal upwelling

D) El Niño

Gravitational circulation is induced by density and elevation differences between freshwater runoff and salt water. Use the following image to answer Questions 57 and 58:



58. (1.00 pts) Which of the following types of circulation mode is this:

A) Classical

B) Reverse

C) Discharge

D) Storage

59. (1.00 pts)
Using $\Delta h = \Delta h(r_2 - r_1)r_1$, where r_1 (freshwater layer density) = 1,000 kg/m^3 | r_2 (saltwater layer density) = 1,075 kg/m^3 | h = depth of equipotential surface at B = 2m, calculate Δ**h**.

A) 0.05 m

B) 0.14 m

C) 0.15 m

D) 5 cm

60. (1.00 pts)
(Tie-breaker) A 75.0 mL wastewater sample was placed into a 400 mL BOD beaker. The rest of the beaker was filled with aerated dilution water. The initial dissolved oxygen was 5.6(mg/L and after 5 days, the dissolved oxygen was 2.40 mg/L. What is the BOD_5 of the water sample:

A) 0.6 mg/L

B) 3.20 mg/L

C) 17.1 mg/L

D) 22.8 mg/L

61. (1.00 pts) The reaction, $N_2 + 3H_2 \rightarrow 2NH_3 \rightarrow$ organic N, refers to which nitrogenous phase:

A) Ammonification
B) Nitrification
C) Denitrification
D) Fixation

Part II: Coral Reef Macroflora & Fauna Identification- Multiple Choice

Questions 62-86

62. (1.00 pts) Which does not affect the spatial pattern of gorgonians?

A) Water depth
B) Light levels
C) Frequency and severity of disturbances
D) Predation by C. gibbosum

63. (1.00 pts) What is the primary role of the parrotfish species in coral reefs?

A) Parrotfish act as important indicators for the health of the coral reef
B) Parrotfish keep algae from overgrowing reef-building corals
C) Parrotfish filter feed small prey from the water above the reef surface
D) Parrotfish help recycle nutrients by breaking down detritus and other organic matter

64. (1.00 pts) Which answer choice is the most true about the long-spined black sea urchin?

A) The species plays a key role in organizing/structuring subtidal rocky reefs
B) The species creates a feedback loop encouraging the growth of disease-causing microbes
C) The skeleton of the sea urchin is composed of calcite and a collagen-like substance
D) The organism can mutilate its own body

65. (1.00 pts) How many larval stages do redbanded coral shrimp have?

A) 3
B) 5
C) 7
D) 9

66. (1.00 pts) What is the symbiotic relationship between zoonxanthellae and coral?

A) Parasitism
B) Amensalism
C) Commensalism
D) Mutualism

67. (1.00 pts) Which of the following marine organisms is an umbrella species:

A) Snapper
B) Humpback wrasse
C) Moral eel
D) Lobster

68. (1.00 pts) What is the life expectancy of crown-of-thorns starfish:

A) 2-3 years
B) 5-8 years
C) 10 years
D) 15 years

69. (1.00 pts) CCA build and cement the carbonate framework of coral reefs as they bind adjacent substrata and provide a calcified tissue barrier against erosion. CCA is:

A) Carbonate constructive algae
B) Crustose calcareous algae
C) Corallitic calcareous algae
D) Calcite conversion algae

70. (1.00 pts) The false eye of *Chaetodon spp.* is located on which bodily region:

A) Above the nostrils
B) Next to the gills
C) Above the pelvic fin
D) Next to the dorsal fins

71. (1.00 pts) (Tie-breaker) Baby butterfly fishes are known as:

A) Fry
B) Hake
C) Sprag
D) Smelt

72. (1.00 pts) Which of the following is a keystone species:

- A) Banded coral shrimp
- B) Grouper
- C) Long-spined black sea urchin
- D) Sponge

73. (1.00 pts) Which of the following marine organisms conducts cooperative hunting with another marine organism:

- A) Snapper
- B) Humphead wrasse
- C) Moray eel
- D) Lobster

74. (1.00 pts) What is the symbiotic relationship between a sea cucumber and a common pearl fish:

- A) Parasitism
- B) Amensalism
- C) Commensalism
- D) Mutualism

75. (1.00 pts) What happens to coral polyps when they don't receive enough sunlight:

- A) The bleaching effect of the sun stops and the coral becomes colorful.
- B) Zooxanthellae leave the coral causing the polyps to turn white.
- C) Nothing since coral polyps aren't living organisms.
- D) Fish and other marine organisms rush to bring the coral back to life.

76. (1.00 pts) A fertilized clam egg floats in the sea for about 12 hours until it eventually hatches as a:

- A) Semaphore
- B) Tridophore
- C) Trochophore
- D) Echinophore

77. (1.00 pts) How many cell division cycles do sea urchins undergo during early embryonic development:

- A) 2
- B) 4
- C) 6
- D) 10

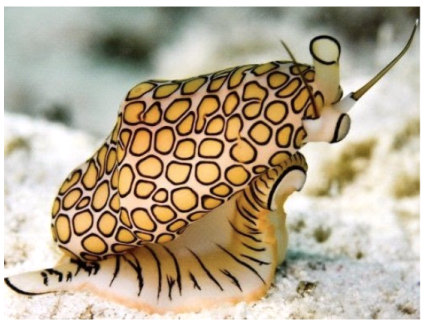
78. (1.00 pts) Aristotle's Lantern was once thought to be:

A) The knotting technique applied by moral eels.
B) The glowing chemicals that make up fleshy algae.
C) A sea urchin's entire chewing organ.
D) A disease exhibited by crown-of-thorns starfish after TCBS exposure.

79. (1.00 pts) Which marine organism participates in aggregated spawning:

A) Snapper
B) Parrotfish
C) Grouper
D) All of the above

80. (1.00 pts) Which of the following marine organism is shown in the image below:



A) *Haemulidae plectorhinchus*
B) *Epinephelus striatus*
C) *Cyphoma gibbosum*
D) *Bolbometopon muricatum*

81. (1.00 pts) A sea cucumber exhibits what type of reproduction:

A) Sexual
B) Asexual
C) A and B
D) None of the above

82. (1.00 pts) (Tie-breaker) The following are possible colors that *Gorgonia* can exhibit except:

A) Purple
B) Yellow
C) Green
D) Red

83. (1.00 pts) A sponge undergoes all of the following reproductive modes except:

A) Sexual reproduction
B) Spore formation
C) Fission
D) Fragmentation

84. (1.00 pts) What is the conservation status on the following marine organism:



A) Least concerned
B) Vulnerable
C) Endangered
D) Near threatened

85. (1.00 pts) Which of the following marine organisms are not found in the Indo-Pacific region:

A) *Tridacna spp.*
B) *Cromileptes altivelis*
C) *Cheilinus undulatus*
D) *Acanthaster planci*

86. (1.00 pts) Sea urchin micromeres form at what relative pole:

A) Oral
B) Aboral
C) Vegetal
D) Animal

Part II: Coral Reef Macroflora & Fauna Identification- Short Answers

Questions 87-107

Use the following image (**Figure A**) to answer the next four questions:

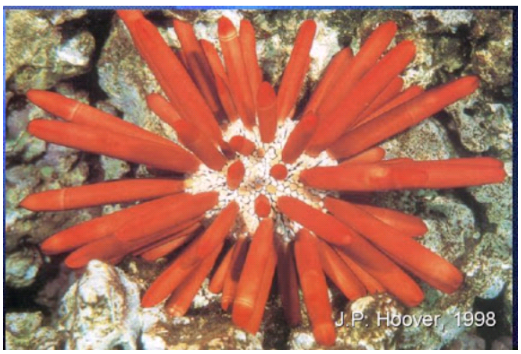


Figure A

87. (1.00 pts) Identify this organism by its common name.

Expected Answer: Red pencil urchin Grading Rubric: +1 for common name

88. (0.50 pts) Identify this organism by its scientific name.

Expected Answer: Heterocentrotus mamillatus Grading rubric: +1 for correct scientific name

89. (1.00 pts) Name the correct family classification of this organism.

Expected Answer: Echinometridae +1 for correct classification

90. (2.00 pts) What environment does this organism live in and what depth are they usually found in?

Expected Answer: reef associated, tropical, depth range = 0-25m Grading Rubric: +0.5 for correctly stating that it's associated with reefs +0.5 for correctly stating that they are found in tropical environments +1 if they mentioned the depth range of either 0-25m or 8-25m

91. (1.00 pts) State in one sentence the mating behavior of this organism.

Expected Answer: members are gonochoric (separate sexes), fertilization is external, spawning Grading Rubric: +1 if they mention the fertilization is external with spawning

Use the following image (**Figure B**) to answer the next three questions:



Figure B

92. (1.00 pts) Identify this organism by its common name.

Expected Answer: Butterfly fish Grading Rubric: +1 if given right common name (butterfly fish)

93. (0.50 pts) Identify the organism by its scientific name.

Expected Answer: Chaetodon lunula Grading rubric: +1 if given correct scientific name (+0.5 if they only identify the genus - Chaetodon)

94. (3.00 pts) Explain why this organism is important to coral reefs.

Expected Answer: butterfly fish are obligate corallivores, meaning they depend on the live tissue of corals for food; because their metabolic/energetic demands are linked to the existence and overall health of the coral substrate, these species of butterfly fish are important candidates for changes in conditions on the coral reef (important indicator species) Grading Rubric: +1 if they mention that butterfly fish are obligate corallivores (or explain what this means) +1 if they linked the fact that these fish are obligate corallivores to their reliance on coral for metabolic existence +1 if they mention that these fish are important indicator species of coral reefs

95. (2.00 pts) Explain in one sentence why the limiting factor you chose for the hypothesis in the previous question makes sense.

Expected Answer: iron is required for the functioning of enzymes that are critical for photosynthesis and nitrogen uptake Grading rubric: Answer may depend on what their hypothesis stated +1 if the reasoning makes sense logically +1 if the reasoning is as stated about (iron required for enzymes)

96. (2.00 pts) What type of reproductive strategy does this organism utilize?

Expected Answer: pair bonds established based on division of labor between mates; butterfly fishes are dusk spawners in evenings around new and full moon; are broadcast spawners so there is no parental care Grading Rubric: +1 if they mentioned that butterfly fishers are broadcast spawners +1 if they mentioned that they are dusk spawners in evenings around new and full moon

Use the following image (**Figure C**) to answer the next five questions:



Figure C

97. (1.00 pts) **Identify Figure C** by its common name.

Expected Answer: +1: Bumphead parrotfish

98. (0.50 pts) **Identify figure C** by its scientific name.

Expected Answer: +1: Bolbometopon muricatum

99. (1.00 pts) **Identify Figure C** by the region it's found in:

Expected Answer: +1: Indo-Pacific region

100. (2.00 pts) **Describe** how bumphead males establish dominance and access to preferred territory that is then used to advertise to females that they are available for mating:

Expected Answer: +2: Bumphead males violently head-butt against each other and the male that wins hovers over a particular spot, maintaining his place in the water column to await the females.

101. (2.00 pts) **Describe** how they significantly contribute to coral reef bioerosion:

Expected Answer: +2: Through consumption. Each adult fish ingests over five tons of structural reef carbonate per year. Note: give full 2 points if they mention consumption of carbonate (ie they don't have to mention how much the fish ingest).

102. (2.00 pts) **Describe** how coral reef bioerosion affects humans in a good and bad way: Provide one good way (1 pt) and one bad way (1 pt).

Expected Answer: +1: Good way- Provides jobs for local communities by offering them opportunities for recreation. +1: Bad way- Coral reefs are considered as a medical cabinet, medical sources to develop treatments for various cancers, human bacterial infections, and viruses.

Use the following image (**Figure D and E**) to answer the next four questions:



Figure D

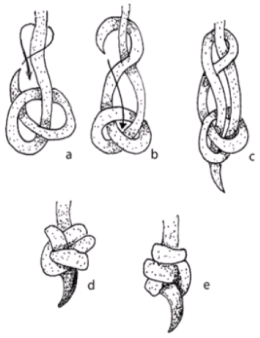


Figure E

103. (1.00 pts) **Identify Figure D** by its common name.

Expected Answer: +1: Moray eel

104. (0.50 pts) **Identify Figure D** by its scientific name.

Expected Answer: +1: Muraenidae

105. (3.00 pts)
Explain why it won't eat the shrimps that come inside of its mouth to eat the parasites living in its mouth. Are the shrimps and that organism communicating in any way? If so, describe how. **What** type of symbiotic relationship does this suggest?

Expected Answer: +1: The mouth acts as a cleaning station for the special shrimps. +1: However, the shrimps must make a “signal” to the eel. Specifically, these shrimps perform a complex dance which notifies the eel that they’re available and ready to clean. +1: Mutualism.

106. (4.00 pts) **What** is the performed technique (**Figure E**) called? **Describe** its purpose and summarize the general process using **Figure E**.

Expected Answer: +1: Knotting. +2: This rotational feeding essentially is where an organism utilizes its longitudinal axis to spin, thereby enabling it to tear apart large prey. ‘Knotting’ can be denoted as a sequence of elaborate movements which include: the positioning of the tail in a backward movement, under the middle of its body, followed by the creation of a second loop by passing the tail back up and into the first loop. The two loops are then tightened, after which the eel draws its head backward through the loops, +1: … thus rendering the prey decapitated, largely torn or swallowed.

107. (1.00 pts) **Identify** what jaws do they use to actively capture and restrain prey.

Expected Answer: +1: Pharyngeal jaws.

Part III: True/False

Questions 108-119

108. (1.00 pts) Photosynthesis is possible in the dysphotic zone

○ True ◉ False

109. (1.00 pts) Estuaries are generally considered ecotones

◉ True ○ False

110. (1.00 pts) Estuary systems with long residence times can export more nutrients to the coastal systems compared to systems with short residence times.

○ True ◉ False

111. (1.00 pts) High oxygen demand and anoxia in estuary systems can be associated with natural processes.

◉ True ○ False

112. (1.00 pts) Mudflats are home to a wide range of organisms that tolerate changing conditions induced by tidal movements.

◉ True ○ False

113. (1.00 pts) Mangroves can act as either a source or a sink of organic matter

◉ True ○ False

114. (1.00 pts) Tropical cyclones and floods can lead to significant nutrient loading (increase in nutrients) for estuaries.

◉ True ○ False

115. (1.00 pts) Conversion of natural forested lands does not affect water quality of adjacent estuaries.

○ True ◉ False

116. (1.00 pts) The San Francisco Bay has large-scale eutrophic/hypoxic conditions due to significant annual nitrogen loads.

○ True ◉ False

117. (1.00 pts) High levels of PCBs in the San Francisco Bay has led to high levels of PCBs in avian eggs.

◉ True ○ False

118. (1.00 pts) C. gibbosum is a generalist browser.

◉ True ○ False

119. (1.00 pts)
Raw sewage water contains **20 mg/L** phosphate, and a secondary sewage treatment plant removed **22%** of the phosphate. A secondary treatment plant would provide potable water given that 0.5 mg/L is the maximum phosphate concentration allowable in potable water.

○ True ◉ False

Part IV: Free Responses

Questions 120-131

120. (4.00 pts) What is the Secchi Disk used for? Explain how the Secchi Disk is used and what its measurement means.

Expected Answer: - to measure water clarity (turbidity) - the Secchi disk is a 8-inch diameter disk with alternating black and white quadrants that is lowered into the water column until it can no longer be seen from the surface - what point the disk disappears is a function of its turbidity (the disk measures the water clarity, and algae/suspended particles makes the water cloudier) - the lower the Secchi depth, the more transparent the water is, the less productive the water is Grading rubric: +1 if they mentioned that it's used for water clarity and/or turbidity +1 if they mentioned what the Secchi disk is (disk lowered into the water column) +1 if they mentioned that the point of disappearance is the turbidity +1 if they mentioned that the lower the depth, the less turbid the water

121. (2.00 pts) The accumulation and deposition of what creates coral reefs? What natural phenomenon is affecting the corals' ability to create these skeletons?

Expected Answer: calcium carbonate/limestone Grading rubric: +1 for calcium carbonate or limestone +1 ocean acidification

122. (3.00 pts) Describe the purpose of dissolved air flotation (DAF) and how it works.

Expected Answer: - used for industrial and municipal processes and wastewater treatment; particularly when suspended particles do not settle down easily in the water by sedimentation - Air on DAF system tanks cause suspended floating mass of required floc → floating floc eliminated from floor - Essential for water w/ algae blooms with low turbidity

and high coloration Grading rubric: +0.5 if they state the purpose of DAF as primarily for wastewater treatment +1.5 if they mentioned it's used when suspended particles… +1 if they mentioned that the air in the tanks lead to floating floc of the particles

123. (3.00 pts)
The osmotic pressure of sea water is approximately 18 atm at 15℃. What is the approximate concentration of salt in sea water (assuming that NaCl has a van't Hoff factor of 2). Use R = 0.082 L*atm/mol*K. Provide the equation you used and the final answer with the correct significant figures.

Expected Answer: osmotic pressure = icRT 18 atm = 2*c*288K*(0.082 L*atm/mol*K) c = 0.38 M Grading Rubric: +1 if they used the right equation +1.5 if they gave the right answer +0.5 if they gave the right sig figs (2)

124. (5.00 pts) List five major threats to coral reefs.

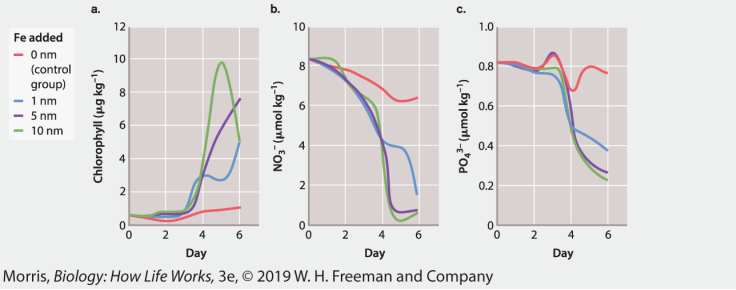
Expected Answer: 1. Elevated ocean temperatures (due to climate change) 2. Increased level of CO2 in the air 3. Pollution from land (including sedimentation, nutrients, pathogens 4. Physical damage/destruction 5. Overfishing 6. Coral harvesting 7. Plastic in coral 8. Dredging 9. Coastal reclamation 10. Offshore drilling Grading Rubric: +1 for each threat they lis

125. (5.00 pts) (Tie-breaker) What beneficial role do viruses play in marine biomes?

Expected Answer: Lyse about 30% of phytoplankton daily in coastal ecosystems, release large amounts of organic molecules (dissolved organic matter) into water column (overlyin the seafloor) → this organic matter supports large populations of respiring bacteria and archaeons that return carbon to environment (in other words, helps with recycling system of the ocean and keeps carbon/nutrients available in surface waters) Grading rubric: +2 if they mentioned that viruses lyse phytoplankton +1 if they mentioned that the lysing of phytoplankton lead to large amounts of organic molecules +2 if they mentioned that the organic matter supports bacteria and archeons

(Tie-breaker) Photosynthetic organisms in the ocean require nitrogen and phosphorous in order to carry out photosynthesis. While the lack of either nitrogen or phosphorous can limit rates of photosynthesis, some regional areas have high concentrations of both. However, primary producers still do not reach maximum primary production, suggesting that some other factor is limiting primary production. (Fe = Iron, NO3- = nitrate, PO4 3- = phosphate)

Use this information to answer the following 3 questions.



Morris, *Biology: How Life Works,* 3e, © 2019 W. H. Freeman and Company

126. (3.00 pts) Come up with a hypothesis for what may be limiting the primary production (use the description).

Expected Answer: iron is the limiting factor Grading rubric: Answers may vary for this question +1 point if the hypothesis makes sense logically +2 points if they mention iron as the limiting factor (max 2 points)

127. (2.00 pts) Explain in one sentence why the limiting factor you chose for the hypothesis makes sense.

Expected Answer: A: iron is required for the functioning of enzymes that are critical for photosynthesis and nitrogen uptake Grading rubric: Answer may depend on what their hypothesis stated +1 if the reasoning makes sense logically +1 if the reasoning is as stated about (iron required for enzymes)

128. (6.00 pts)
The experiment that was carried out included taking samples of seawater with photosynthesis phytoplankton and adding small amounts of iron. Daily measurements were taken, and the results are shown in the graph below. Explain in one to three sentences what conclusions you can draw from the results/graph.

Expected Answer: Addition of iron increased rates of photosynthesis, which allowed primary producers to take up more nitrogen and phosphorus. This supports the idea that iron limits primary production. Grading rubric: Answer may depend on interpretation +2 if they mentioned that iron increases rates of photosynthesis +2 if they mentioned that increased iron leads to higher intake of nitrogen and phosphorus +2 if they come to the conclusion that iron limits primary production

129. (3.00 pts)
Name three different types of potent toxins/poisoning that marine species can produce which can cause gastrointestinal and neurological illnesses in humans.

Expected Answer: Paralytic shellfish poisoning (PSP), diarrhetic shellfish poisoning (DSP), amnesic shellfish poisoning (ASP), ciguatera fish poisoning (CFP), neurotoxic shellfish poisoning (NSP), cyanobacterial toxin poisoning, estuarine associated syndrome Grading Rubric: +1 point for each toxin they name

130. (2.00 pts)
Non-native species are often introduced to estuaries in the ballast water of ships. When ships are empty, they take in water to help keep them balanced. When cargo is loaded onto the ships, they release the ballast water. In addition to water, aquatic organisms are sucked into the ships' ballast tanks. One of these organisms are zebra mussels. In regards to population dynamics, let's say that the number of zebra mussels in a river grows at an exponential growth rate of **450%** per year. **Round your final answer to two significant figures**.

Expected Answer: +2: 0.46 years P(t) = Po (a)^t | 2Po = P(t); a = 4.50

131. (2.00 pts) **Give** two forever chemicals that bring environmental concerns to marine and human life:

Expected Answer: +2: (+1 each) Fluorinated chemicals: Perfluorooctanesulfonic acid (PFOS), Perfluroroctanoic acid (PFOA), Perfluoronoamnoic acid (PFNA), Perfluorobutanesulfonic acid (PFBS), and Perfluorohexanoic acid (PFHxA)

Congratulations! You survived this very long test about water and the ocean and estuaries and other random stuff. We know you gave it your all, and hopefully it was a (semi) fun test to take :))



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